MANIPAL INSTITUTE OF TECHNOLOGY
MANIPAL
(A constituent unit of MAHE, Manipal)

DEPARTMENT OF CIVIL ENGINEERING

7TH SEMESTER B.TECH (CIVIL ENGINEERING)

END SEMESTER EXAMINATION, NOV 2024

PAVEMENT MATERIALS AND DESIGN (CIE 4069)

(8 - 12 - 2024)

TIME: 3 HRS. **MAX.MARKS: 50**

Note: 1. Answer all questions.
2. Any missing data may be suitably assumed.
3. Use of Formula book, Design charts and IRC codes are permitted

Q. No.	Question	Marks	NBA		BL
			CO	PO	
1A	In the Marshall method of mix design, the coarse aggregates, fine aggregates, filler and bitumen, having respective specific gravities of 2.62, 2.72, 2.70 and 1.02 are mixed in the ratio of 55, 34.6, 4.8 and 5.6%, respectively. Calculate the theoretical specific gravity of the mix. If the actual specific gravity is 2.2, determine the percentage of air voids.	**5**	2		**4**
1B	With the help of a neat graph, illustrate the relationship between bitumen content and Marshall stability.	**3**	2		**3**
1C	Discuss the role of fillers in a bituminous mix.	**2**	2		**2**
2A	A three layer pavement structure consists of 300 mm of bituminous layer on 150 mm of crushed stone road base. The subgrade has a CBR of 5. A single wheel load with a tyre pressure of 0.5 MN/m² is applied over a circular contact area of 120 mm radius. Determine the vertical and horizontal stresses and strains on both sides of each interface. E1 = 2000 MN/m2, E2 = 100 MN/m2, and E3 = 50 MN/m2 are the modulus of elasticities of the bituminous layer, granular layer, and subgrade, respectively.	**5**	3		**3**
2B	Discuss any three factors affecting the mechanical stabilization of soil.	**3**	5		**2**
2C	List out the possible causes of raveling with the help of a neat figure.	**2**	5		**2**

<table>
<tr><td>3A</td><td>Design size and spacing of dowel bars at an expansion joint of concrete pavement of thickness 25 cm. The load capacity of the dowel system is 40 percent of the design wheel load. The joint width is 2.0 cm, and the permissible stress in shear, bending, and bearing stress in dowel bars are 1000, 1400, and 100 kg/cm2, respectively. Consider the radius of relative stiffness to be 80 cm and design wheel load as 5000 kg.</td><td>5</td><td>3</td><td></td><td>5</td></tr>
<tr><td>3B</td><td>Distinguish between falling weight deflectometer and ground penetrating radar methods for structural evaluation of flexible pavements.</td><td>3</td><td>5</td><td></td><td>2</td></tr>
<tr><td>3C</td><td>List out the properties of the bituminous mix.</td><td>2</td><td>2</td><td></td><td>2</td></tr>
<tr><td>4A</td><td>Discuss in detail the process of determination of modulus of subgrade reaction. Explain the significance of the same in the design of rigid pavements.</td><td>5</td><td>2</td><td></td><td>2</td></tr>
<tr><td>4B</td><td>
<table>
<tr><th>Bitumen content %</th><th>Density g/cc</th><th>Air voids (%)</th><th>VMA (%)</th><th>VFB (%)</th><th>Stability (in Kn)</th><th>Flow in mm</th></tr>
<tr><td>4.00</td><td>2.279</td><td>8.10</td><td>16.48</td><td>50.86</td><td>1060</td><td>1.5</td></tr>
<tr><td>4.50</td><td>2.310</td><td>6.19</td><td>15.78</td><td>60.80</td><td>1205</td><td>2.0</td></tr>
<tr><td>5.00</td><td>2.320</td><td>5.15</td><td>15.88</td><td>67.61</td><td>1252</td><td>2.8</td></tr>
<tr><td>5.50</td><td>2.338</td><td>3.72</td><td>15.66</td><td>76.25</td><td>1434</td><td>3.2</td></tr>
<tr><td>6.00</td><td>2.316</td><td>3.97</td><td>16.46</td><td>75.86</td><td>1183</td><td>3.9</td></tr>
<tr><td>6.50</td><td>2.302</td><td>3.91</td><td>16.98</td><td>76.95</td><td>1038</td><td>4.3</td></tr>
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The results of Marshall’s stability test are given above. Estimate the optimum binder content.
</td><td>3</td><td>2</td><td></td><td>5</td></tr>
<tr><td>4C</td><td>Discuss the factors affecting soil stabilization using cement.</td><td>2</td><td>5</td><td></td><td>2</td></tr>
<tr><td>5A</td><td>With the help of a neat figure, explain the Benkelman Beam Deflection method for structural evaluation of the pavement.</td><td>5</td><td>5</td><td></td><td>2</td></tr>
<tr><td>5B</td><td>Elaborate on the factors defined for the design of runway pavements by the Federal Aviation Agency method.</td><td>3</td><td>4</td><td></td><td>3</td></tr>
<tr><td>5C</td><td>Discuss the merits and demerits of mathematical and trial-and-error methods for solving the problem of blending aggregates.</td><td>2</td><td>2</td><td></td><td>2</td></tr>
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